

Overview of Headache

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A headache is pain in any part of the head, including the scalp, upper neck, face, and interior of the head. Headaches are one of the most common reasons people visit a doctor.

Headaches interfere with the ability to work and do daily tasks. Some people have frequent headaches. Other people hardly ever have them.

Causes of Headache

Although headaches can be painful and distressing, they are rarely due to a serious condition. Headaches can be divided into 2 types:

- **Primary headaches:** Not caused by another disorder
- **Secondary headaches:** Caused by another disorder

Primary headache disorders include

- [Tension-type headache](#)
- [Migraine](#)
- [Cluster headache](#) and other trigeminal autonomic cephalgias, including chronic paroxysmal hemicrania, hemicrania continua, and [short-lasting unilateral neuralgiform headache with conjunctival injection and tearing](#) (SUNCT)

Trigeminal autonomic cephalgias are rare. (*Cephalgia* is the medical term for headache.)

Secondary headaches may result from disorders of the brain, eyes, nose, throat, sinuses, teeth, jaws, ears, or neck or from a bodywide (systemic) disorder.

Common headaches

The **2 most common types of headache** are primary headaches:

- [Tension-type](#) (most common overall)
- [Migraine](#)

Less common headaches

Less often, headaches are due to a less common primary headache disorder called [cluster headache](#) or to one of the many secondary headache disorders (see table [Some Causes and Features of Headaches](#)). Some secondary headache disorders are serious, particularly those that involve the brain, such as [meningitis](#), a [brain tumor](#), or bleeding within the brain ([intracerebral hemorrhage](#)).

Fever can cause headaches, as can many infections that do not specifically involve the brain. Such infections include [Lyme disease](#), [Rocky Mountain spotted fever](#), and [influenza](#).

Headaches also commonly occur when people stop consuming [caffeine](#) or stop taking pain relievers (analgesics) after using them for a long time (called [medication overuse headache](#)).

Contrary to what most people think, eye strain and high blood pressure (except for extremely high blood pressure) do not typically cause headaches.

Evaluation of Headache

Doctors focus on the following:

- Determining whether the headache has another cause (that is, whether it is a secondary headache)
- Checking for symptoms suggesting that the headache is caused by a serious disorder

If no cause is identified, they focus on identifying which type of primary headache is present.

Warning signs

In people with headaches, certain characteristics are cause for concern:

- Changes in sensation or vision, sudden weakness, loss of coordination, seizures, difficulty speaking or understanding speech, or changes in levels of consciousness such as drowsiness or confusion (suggesting a brain disorder)
- A fever and a stiff neck that makes lowering the chin to the chest painful and sometimes impossible
- A very sudden, severe headache (thunderclap headache)
- Tenderness at the temple (as when combing hair) or jaw pain when chewing
- The presence of cancer or a disorder that weakens the immune system ([immunodeficiency disorder](#)), such as [advanced HIV infection](#)
- Use of a [medication that suppresses the immune system](#) (immunosuppressant)
- Symptoms that affect the whole body such as fever or weight loss
- A headache that progressively worsens (in frequency or severity)
- Red eyes and halos seen around lights
- Very high blood pressure
- Headaches that begin after age 50

When to see a doctor

People who have any warning sign should see a doctor immediately. The presence of a warning sign may suggest that the headaches may be caused by a serious disorder, such as one of the following:

- A severe headache with a fever and a stiff neck: [Meningitis](#)—a life-threatening infection of the fluid-filled space between the tissues covering the brain and spinal cord (meninges)
- A thunderclap headache: [Subarachnoid hemorrhage](#) (bleeding within the meninges), which is often due to a ruptured aneurysm
- Tenderness at the temple, particularly in older adults who have lost weight and have muscle aches: [Giant cell arteritis](#)
- Headaches in people who have cancer or a weakened immune system (due to a disorder or medication): Meningitis or spread of cancer to the brain

- Red eyes and halos seen around lights: [Glaucoma](#), which, if untreated, leads to irreversible loss of vision

If people with none of the above symptoms or characteristics start having headaches that are different from any they have had before, or if their usual headaches become unusually severe, they should call their doctor. Depending on their other symptoms, the doctor may ask them to come for an evaluation.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the pain and tests that may need to be done (see table [Some Causes and Features of Headaches](#)).

Doctors ask about the characteristics of the headache:

- How often it occurs
- How long it lasts
- Where the pain is
- How severe is it
- What the pain is like (for example, whether it is throbbing, dull, or like a knife)
- Whether any symptoms accompany it
- How long a sudden headache takes to reach its maximum intensity
- What triggers the headache (for example, whether it occurs only when standing), what makes it worse, and what relieves it

Other questions may include

- Whether people have had headaches before
- Whether the headaches recur and, if so, when did they start and how often do they occur
- Whether the current headache is the same or different from previous headaches

Doctors also ask about risk factors for headache. They include

- Whether people take or have stopped taking certain medications or substances (particularly ingesting too much or stopping [caffeine](#))
- Whether they have a disorder that may account for the headache
- Whether they have family members with severe headaches

- Whether they have had a recent head injury
- Whether they have had a spinal tap recently

People can think about how to answer the above questions and write the answers down before they go to the doctor. Sometimes doctors ask people to fill out a headache questionnaire that covers most of the relevant questions. People may complete the questionnaire before their visit and bring the results with them. Having this information written down can save time and help guide the evaluation.

A general physical examination is done. It focuses on the head and neck and on brain, spinal cord, and nerve function ([neurologic examination](#)). An [eye examination](#) is sometimes also done.

Testing

Most people do not need testing. However, if doctors suspect a serious disorder, tests are done. For some suspected disorders, tests are done as soon as possible. In other cases, testing can be done within one or more days.

Magnetic resonance imaging (MRI) and/or **magnetic resonance angiography** (which provides detailed images of blood vessels) or, if MRI is not available or contraindicated, **computed tomography (CT)** is done as soon as possible if people have

- A thunderclap headache
- Changes in levels of consciousness, such as drowsiness or confusion
- A fever and a stiff neck that makes lowering the chin to the chest painful and sometimes impossible
- Swelling of the optic nerve ([papilledema](#)), detected by eye examination with an ophthalmoscope
- Symptoms that suggest a serious bodywide response to an infection ([sepsis](#)), such as a certain type of rash or shock
- Symptoms that suggest a brain disorder, such as changes in sensation or vision (including double vision), sudden weakness, loss of coordination, seizures, or difficulty speaking or understanding speech
- A head injury causing headache and loss of consciousness

If people have a thunderclap headache, [magnetic resonance angiography](#) (or CT) is done immediately.

MRI is done within a day or so if people have conditions such as the following:

- Cancer

- A weakened immune system (due to a disorder such as HIV infection or a medication)

MRI or CT (if MRI is not available or is contraindicated) is done within a few days if people have certain other characteristics, such as the following:

- Headaches that begin after age 50
- Double vision
- A new headache that is worse when the person awakens in the morning or that awakens the person from sleep
- An increase in the frequency, duration, or intensity of chronic headaches

A **spinal tap** (lumbar puncture) is usually done if

- Acute [meningitis](#) or [encephalitis](#) (a brain infection) is suspected.
- People have a thunderclap headache (suggesting [subarachnoid hemorrhage](#)) even when results of CT or MRI or other imaging tests are normal.
- People have a weakened immune system.
- The headache is worsening and/or doctors suspect [idiopathic intracranial hypertension](#) or [chronic meningitis](#).

Usually, doctors do CT or MRI before the spinal tap if they think that pressure within the skull may be increased—for example, by a mass (such as a tumor, an abscess, or a hematoma). A spinal tap can be dangerous if pressure within the skull is increased. When spinal fluid is removed and pressure within the skull is increased, parts of the brain may suddenly shift downward. If these parts are pressed through the small openings in the tissues that separate the brain into compartments, a life-threatening disorder called [brain herniation](#) results.

Other tests are done within hours or days, depending on the examination results and the causes that are suspected.



Some Causes and Features of Headaches

Type or Cause	Common Features*	Diagnostic Approach†
Primary headache (not due to another disorder)	A severe, piercing headache that • Affects one side of the head and is focused around the eye • Lasts 30 to 180 minutes (usually 30 minutes to 1 hour) • Often occurs at the same time of day • Occurs in clusters, separated by periods of time when no headaches occur • Is usually not worsened by light, sounds, or odors • Is not accompanied by vomiting	A doctor's examination alone Occasionally MRI of the head to rule out other disorders, particularly if the headaches have developed recently or if the pattern of symptoms has changed
<u>Cluster headache</u>	Inability to lie down and restlessness (sometimes expressed by pacing) On the same side as the pain: A runny nose, tearing, drooping of the eyelid	